

Sleep Quality Predictor

PROJECT DESCRIPTION:

Sleep is essential for physical and mental well-being, and poor sleep quality is linked to stress, obesity, heart disease, and reduced productivity. This project aims to develop a **Machine Learning-based Sleep Quality Predictor** that analyzes lifestyle, behavioral, and physiological factors to predict the quality of a person's sleep.

The model uses data such as screen time, exercise, caffeine intake, stress levels, and sleep duration to classify sleep quality as **Good**, **Average**, or **Poor**. The goal is to provide users with actionable insights to improve their sleep health and overall lifestyle.

FEATURES:

- Predicts sleep quality based on daily habits and health data
- User inputs include screen time, physical activity, bedtime, stress level, etc.
- Uses supervised ML algorithms for prediction
- Provides personalized tips for improving sleep
- Simple interface for entering daily data and viewing predictions
- Can be extended into a mobile or web-based wellness tool

TECHNOLOGIES USED:

- **Language:** Python
- **Libraries:**
 - scikit-learn – for model training and evaluation
 - pandas, NumPy – for data preprocessing
 - Matplotlib, Seaborn – for visualizations

- **Algorithms:** Logistic Regression, Random Forest, Decision Tree, SVM
- **NLP (Optional):** For analyzing user-entered sleep diary or comments
- **Dataset:**
 - Sleep Health and Lifestyle Dataset (Kaggle)
 - Sleep Efficiency Dataset (UCI ML Repo)

OUTCOMES:

- Built a predictive model to evaluate sleep quality based on real-world health/lifestyle data
- Helped identify key factors affecting poor sleep
- Encouraged users to track and improve sleep-related habits
- Laid the foundation for future smart sleep apps and health monitoring tools
- Enhanced understanding of behavioral data modeling and health analytics

User Interface Layout: Sleep Quality Predictor (Example)

Sleep Quality Predictor

A smart way to monitor and improve your sleep

Input Section:

Input Field	Type	Description
Sleep Duration (in hours)	Numeric Input	Total hours you slept last night
Bedtime	Time Picker	Time you went to bed
Wake-up Time	Time Picker	Time you woke up
Caffeine Intake	Dropdown	None / Low / Moderate / High

Input Field	Type	Description
Exercise Duration	Numeric Input	Minutes of exercise done during the day
Screen Time Before Bed	Numeric Input	In minutes (e.g., phone/laptop usage)
Stress Level	Slider (0–10)	Rate your stress level for the day
Mood Before Sleep	Dropdown	Happy / Neutral / Sad / Anxious
Sleep Interruptions (Yes/No)	Radio Button	Woke up during the night?

Buttons & Controls:

- **Predict Sleep Quality** (*Main Action Button*)
- **Reset Inputs**
- **View Past Predictions** (optional)

Output Section: (Example)

- **Predicted Sleep Quality:** *Good / Average / Poor*
- **Suggestion/Tips:**
“Try reducing screen time by 30 minutes before bed.”
“Increase your exercise duration for better sleep.”

Optional: Track History

- Table or chart showing past sleep scores, trends, and improvement tips.

Design Style Suggestions:

- **Colors:** Use soft blues and purples for a calm sleep-related aesthetic
- **Icons:** Use moon, bed, clock icons for better user experience

- **Font:** Clear and readable font like Poppins or Roboto
- **Responsive Layout:** For desktop and mobile compatibility